

Dr. Himanshu Kumar Dwivedi

Curriculum Vitae

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Department of Mathematical Sciences, IIT(BHU) Varanasi, India 221005

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I am a recent Ph.D. graduate from Mathematical Sciences at IIT (BHU), Varanasi, working at the interface of numerical analysis, scientific computing, and scientific machine learning for partial differential equations. My research focuses on the development of fast, high-order, and theoretically rigorous numerical methods for fractional and tempered fractional PDEs, with particular emphasis on accelerated algorithms, nonlocal operators, stability and convergence analysis, and hybrid machine learning-based computational frameworks. I have experience in designing and implementing advanced numerical schemes in Python and MATLAB, and my work has resulted in publications in leading journals in scientific computing. I am seeking a postdoctoral position where I can further develop mathematically grounded computational methods for challenging PDE, integral-equation, and multiphysics problems. Good communication and management skills developed through a working experience in different projects and with different responsibilities, in multidisciplinary teams with members of different nationalities. Indian mother-tongue and proficient in English.

Personal information

Date of birth: July 25th, 1998

Birthplace: Varanasi, India **Nationality:** Indian

Research Interests

Numerical Solution of PDEs; Finite Difference Method; Scientific Machine Learning(PINNs); Finite Element Methods

Education

PhD in Mathematics (9.45 CGPA)

Department of Mathematical Sciences, Indian Institute of Technology (BHU) Varanasi, India

Jan. 2022–Apr. 2026

Master degree in Mathematics(8.22 CGPA)

Indian Institute of Technology Bhilai, India

Jul. 2019–June 2021

Bachelor degree in Mathematics

Mahatma Gandhi Kashi Vidyapith University(Udai Pratap College), India

Jul. 2015–June 2018

PhD thesis (Defense date: 31 July 2026)

Title. Analysis of Accelerated Finite-Difference and Scientific Machine Learning Schemes for Time-Fractional Partial Differential Equations

Institution. Indian Institute of Technology (BHU) Varanasi, India

Advisor. Prof. Rajeev

Summary. In this thesis, we develop accurate and computationally efficient numerical methods for nonlinear and multidimensional fractional PDEs. The work focuses on accelerated discretizations for Caputo and tempered Caputo derivatives, high-order compact finite difference schemes, nonuniform time-stepping for weakly singular solutions, and rigorous stability and convergence analysis. It also integrates these numerical frameworks with scientific machine learning methods such as PINNs, XfPINNs, and DeepONet. The proposed methods significantly reduce memory and computational cost while preserving high accuracy and strong theoretical guarantees. The thesis addresses applications including Burgers equations, subdiffusion models, nonlinear Schrödinger equations, and phase-field equations such as Allen-Cahn.

Link. [Download PDF](#)

Master's thesis

Title. A Study of Certain Subset Sum Problems Through a Unified Approach

Institution. Indian Institute of Technology Bhilai, India

Advisor. Prof. Raj Kumar Mistri

Summary. In this thesis, we investigate direct and inverse problems for subset sums in additive abelian groups by expressing them within the framework of general h-fold sumsets. It established that both classical subset sums and restricted subset sums can be represented as special cases of generalized sumsets, thereby providing a unified perspective on these problems. As a consequence, the thesis obtained alternate proofs of several known results in the additive group of integers, including direct and inverse theorems due to Nathanson and later authors. The work lies in additive combinatorics and highlights the usefulness of unified sumset formulations in studying subset sum structures

Personal skills

Languages

Hindi: mother tongue

English: level C1

	Written comprehension	Writing	Listening	Speaking
Hindi	Mother tongue	Mother tongue	Mother tongue	Mother tongue
English	Professional	Professional	Professional	Professional

Computer skills

Operating systems: Ubuntu, Windows, MacOS

Computational software: FreeFEM

Programming languages: Matlab, Python, C, Mathematica

Text editing: L^AT_EX, Microsoft Office

Publications

Papers published

1. S. Kumar, **H.K. Dwivedi***, Rajeev, *A deep learning WKB-PINNs framework for forward and inverse problems of semiclassical nonlinear Schrödinger equations* Commun. Nonlinear Sci. Numer. Simul.(Accepted)
2. S. Kumar, **H.K. Dwivedi***, Rajeev, *Accelerated singularity-aware fractional PINNs for time-fractional phase-field equations: forward and inverse problems.* J. Comput. Sci., 100, 102969, (2026) ([link](#))
3. S. Kumar, **H.K. Dwivedi**, M. Ehrhardt and Rajeev, *A Fractional-Memory Physics-Informed Neural Network with Fast History Compression for Tempered Fractional Coupled Phase-Field Systems* arXiv (2026) arXiv.2606.22191 ([link](#))
4. S.Kumar, **H.K. Dwivedi***, Rajeev, *A novel time-discrete physics-informed neural network framework for forward and inverse problems of the 2D tempered fractional generalized Burgers equation.* Comput. Math. Appl. 219, 85-109 (2026) ([link](#))
5. **H.K. Dwivedi*** and Rajeev, *An adaptive finite difference/ ADI scheme with maximum principle preservation for 2D Allen-Cahn equation with accelerated Caputo derivative evaluation,* Comput. Math. Appl. 217, 402-429, (2026) ([link](#))
6. **H.K. Dwivedi**, M. Ehrhardt and Rajeev, *Alikhanov-XfPINNs: Adaptive physics-informed learning for nonlinear fractional PDEs on nonuniform meshes,* arXiv (2026) arXiv:2605.01305 ([link](#))
7. G. Su, **H.K. Dwivedi** and J. Pan, *An accelerated Alikhanov-DeepONet operator learning framework for time-fractional phase-field models,* J. Comput. Appl. Math., Vol. 487 (2026), p. 117711 ([link](#))
8. S. Kumar, **H.K. Dwivedi*** and Rajeev, *Structure preserving conservative phase-amplitude PINNs for semiclassical Schrödinger-type equations,* J. Comput. Appl. Math., Vol. 485 (2026), p. 117607 ([link](#))
9. **H.K. Dwivedi**, Rajeev and S. Zeng, *Nonuniform tempered Alikhanov scheme for fractional Allen-Cahn equations with discrete maximum principle,* J. Sci. Comput., Vol. 107(1) (2026), p. 17 ([link](#))

10. **H.K. Dwivedi**, Y. Peng, Rajeev and S. Zeng, *Finite difference/deep learning scheme for tempered fractional generalized Burgers' equations with fast evaluation of Caputo derivative*, Appl. Math. Comput., Vol. 523 (2026), p. 130032 ([link](#))
11. **H.K. Dwivedi**, Rajeev and S. Zeng, *An alternating direction implicit method for 2D nonlinear Schrödinger equation with accelerated evaluation of Caputo derivative*, Appl. Math. Comput., Vol. 518 (2026), p. 129905 ([link](#))
12. **H.K. Dwivedi*** and Rajeev, *A novel accelerated tempered algorithm with nonuniform time-stepping compact ADI scheme for 2D tempered-fractional nonlinear Schrödinger equations with weak initial singularity*, Comput. Math. Appl., Vol. 196 (2025), pp. 312-337 ([link](#))
13. **H.K. Dwivedi*** and Rajeev, *A novel fast tempered algorithm with high-accuracy scheme for 2D tempered-fractional reaction-advection-subdiffusion equation*, Comput. Math. Appl., Vol. 176 (2024), pp. 371-397 ([link](#))
14. **H.K. Dwivedi** and Rajeev, *A novel fast second-order approach with high-order compact difference scheme and its analysis for tempered fractional Burgers equation*, Math. Comput. Simul., Vol. 227 (2024), pp. 168-188 ([link](#))
15. **H.K. Dwivedi** and Rajeev, *A fast difference scheme for the multi-term time-fractional advection-diffusion equation with nonlinear source term*, Chin. J. Phys., Vol. 89 (2024), pp. 86-103 ([link](#))
16. **H.K. Dwivedi*** and Rajeev, *Fast high-order linearized exponential methods for efficient simulation of 2D time-fractional Burgers equation in polymer solution dynamics*, J. Math. Chem., Vol. 63 (2024), pp. 596-625 ([link](#))
17. **H.K. Dwivedi** and R.K. Mistri, *Direct and inverse problems for subset sums with certain restriction*, Integers: Electronic J. Combinatorial Number Theory, Vol. 22 (2022), pp. 1-13 ([link](#))

Conference Proceedings

1. H.K. Dwivedi and Rajeev, *A new tempered Alikhanov formula with nonuniform time-stepping maximum principle preserving scheme for tempered fractional Allen-Cahn equation*. SIAM Conference on Analysis of Partial Differential Equations (SIAM PD'25), (2025)

Papers submitted

1. **H.K. Dwivedi**, Y. Peng, Rajeev and X. Li, *Meshless operator learning framework and nonuniform time-stepping schemes for subdiffusion problems*, revised in: Commun. Nonlinear Sci. Numer. Simul. (2026)
2. **H.K. Dwivedi** and Rajeev, *An adaptive finite difference/ADI scheme with maximum principle preservation for 2D Allen-Cahn equation with accelerated Caputo derivative evaluation*, revised in Comput. Math. Appl. (2026)
3. **H.K. Dwivedi**, *Finite difference/finite element scheme for time-fractional dispersion equation with accelerated evaluation of Caputo derivative*, Under Review.
4. **H.K. Dwivedi**, Rajeev and S. Zeng, *A novel accelerated finite difference scheme for the generalized tempered fractional Burgers model of porous-media wave propagation*, Under Review.
5. S. Kumar, **H.K. Dwivedi** and Rajeev, *A deep learning WKB-PINNs framework for forward and inverse problems of Schrödinger equation*, Under Review.
6. S. Kumar, **H.K. Dwivedi** and Rajeev, *The WKB-PINNs framework for forward and inverse problems of semiclassical nonlinear Schrödinger equation*, Under Review.
7. **H.K. Dwivedi**, A.S. Hendy and A.A. Alikhanov, *The discrete maximum principle and sharp convergence analysis for a fast scheme of the tempered time-fractional Allen-Cahn equation*, Under review.

Citation scores

Scopus: h-index 4, 59 citations

 **Google Scholar:** h-index 6, i-10 index 3, 82 citations

 **Research Gate:** h-index 4, RG Score 53.4, 59 citations

Teaching Assistance

Academic courses

Finite Element Analysis(MA424)

Bachelor degree in Engineering, IIT(BHU) Varanasi, India

Jan. 2026–May 2026

Engineering Mathematics(MA101)

Bachelor degree in Engineering, IIT(BHU) Varanasi, India

Jul. 2025–Nov. 2025

Finite Element Analysis(MA424)

Bachelor degree in Engineering, IIT(BHU) Varanasi, India

Jan. 2025–May 2025

Numerical Techniques(MA201)

Bachelor degree in Engineering, IIT(BHU) Varanasi, India

Jul. 2024–Nov. 2024

Finite Element Analysis(MA424)

Bachelor degree in Engineering, IIT(BHU) Varanasi, India

Jan. 2024–May 2024

Numerical Techniques(MA201)

Bachelor degree in Engineering, IIT(BHU) Varanasi, India

Jul. 2023–Nov. 2023

Visiting research appointments

Chongqing Normal University, Chongqing, China

Invited by Prof. Shengda Zeng

Jan. 2026

Yulin Normal University, Yulin, Guangxi, China

Invited by Prof. Shengda Zeng

Jan. 2026

Talks

A fast high-order compact scheme for the tempered fractional Burgers equation.

“Yulin Normal University”

Yulin, Gungxi, China

Jan. 9, 2026

Accelerated second-order scheme for the tempered fractional Burgers equation.

Conference “Ramanujan Mathematical Society (RMS)”

Christ University, Bangalore, India

Dec., 2024

A high-order numerical scheme for time-fractional Burgers equation

Conference “Recent Advances in Applied Mathematics(RAAM)”

IIT(BHU) Varanasi, India

Jul. 22, 2024

Additional academic activities and merits

Reviewer for JCR indexed journals

- Applied Mathematics and computation(Elsevier)
- Numerical Algorithms(Springer)
- Mathematics and Computers in Simulation(Elsevier)
- Computers and Mathematics with Applications(Elsevier)
- Fractional Calculus and Applied Analysis(Springer)
- Communications in Nonlinear Science and Numerical Simulation(Elsevier)
- Journal of Computational and Applied Mathematics(Elsevier)
-

Grants

ANRF, SERB(Government of India): *International Travel Grant for attending **SIAM’s** Conference on Analysis of Partial Differential Equations(PD 25), Pennsylvania, USA (15 November, 2025 to 21 November, 2025) 3.0 Lacs INR*

IIT(BHU),Varanasi, India: *Travel Grant for a research visit to Yulin Normal University, Guangxi, China and Chongqing Normal University, Chongqing China(04 January to 03 February, 2026): 1.5 Lacs INR*

References

Prof. Rajeev (rajeev.apm@iitbhu.ac.in)

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Prof. Matthias Ehrhardt (ehrhhardt@uni-wuppertal.de)

Professor for Applied and Computational Mathematics, University of Wuppertal, Germany

Prof. Ahmed S. Hendy (ahmed.hendy@fsc.bu.edu.eg)

Department of Computational Mathematics and Computer Science, Institute of Natural Sciences and Mathematics, Ural Federal University, Russia

Prof. Shengda Zeng (zengshengda@163.com)

Faculty of Mathematics and Computer Sciences, Jagiellonian University, Krakow, Poland

Prof. Subir Das (sdas.apm@iitbhu.ac.in)

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